Linear Algebra Review

Eigenvalues and Eigenvectors

A square matrix M : has eigenvector v and eigenvalue λ if:

$$Mv = \lambda v$$

Ex: $M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -2 \end{pmatrix}$ then $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ e-vector with e-val $= 1$

$v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ " " $= 3$

$v_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ " " $= -2$

We can compute e-values by computing the roots of:

$$\det(\lambda I - M)$$

Linear Algebra Review

symmetric matrices, eigenvalues and eigenvectors (1/2)

M is symmetric if $M_{ij} = M_{ji} \ \forall \ i,j$: $M = M^T$

M symmetric $\Rightarrow$ eigenvalues are <u>real</u>.

$$M = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \qquad \det(\lambda I - M) = \det\begin{pmatrix} \lambda & -1 \\ 1 & \lambda \end{pmatrix} = \lambda^2 + 1.$$

not symmetric

$$M = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \qquad \det(\lambda I - M) = \lambda^2 - 6\lambda + 8$$

$$= (\lambda - 4)(\lambda - 2)$$

symmetric

e-values : 4, 2

Linear Algebra Review

symmetric matrices, eigenvalues and eigenvectors (2/2)

More is true: if M is symmetric, then its e-vectors are perpendicular to each other. (*)

$$M = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \qquad v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$M v_1 = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 4 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

v_1 an e-vec w/ e-val $= 4$

$$M v_2 = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 2 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

v_2 e-vec w/ e-val $= 2$.

$$v_1 \perp v_2 : \quad \begin{pmatrix} 1 \\ -1 \end{pmatrix}^T \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0.$$

Can always write:

$$M = \sum \lambda_i \underline{\underline{v_i v_i^T}}$$

matrix

outer product

Linear Algebra Review

positive semidefinite matrices (1/3)

$M = M^T$ is called PSD if

$\lambda_i \geq 0$.

So: $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ is psd.

Linear Algebra Review

positive semidefinite matrices (2/3)
A is psd iff the quadratic it defines is nonnegative

M is called PSD <u>iff</u> $x^T M x \geq 0 \quad \forall x \in \mathbb{R}^n$

$$(x_1 \ x_2)\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \geq 0 \quad \forall x_1, x_2 \in \mathbb{R}$$

Linear Algebra Review

positive semidefinite matrices (3/3)
for A any matrix, AA' and A'A are always psd

$$x^T A^T A x = \underbrace{(Ax)^T (Ax)}_{u^T u = \|u\|_2^2} = \|Ax\|_2^2 \geq 0$$

Eigenvalue Decomposition for Symmetric Matrices

$M = M^T$ <u>then</u>

$$M = \sum \lambda_i v_i v_i^T$$

where v_i is the (normalized) eigenvector of M with eigenvalue λ_i.

If M is psd, then $\lambda_i \geq 0$.

Check: $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} = 4 \cdot \left(\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right)\left(\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right)^T + 2\left(\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right)\left(\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right)^T.$